

Question	Marking Guidance	Mark	Comments																													
09.1	4 rows correct = 2 marks;; 2 or 3 rows correct = 1 mark; 0 or 1 row correct = 0 marks	2																														
	<table><tr><th rowspan="2">Biological molecules</th><th colspan="4">Element</th></tr><tr><th>Carbon</th><th>Nitrogen</th><th>Oxygen</th><th>Phosphorous</th></tr><tr><td>Galactose</td><td>✓</td><td></td><td>✓</td><td></td></tr><tr><td>Phospholipid</td><td>✓</td><td></td><td>✓</td><td>✓</td></tr><tr><td>RNA</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td></tr><tr><td>Sucrose</td><td>✓</td><td></td><td>✓</td><td></td></tr></table>	Biological molecules	Element				Carbon	Nitrogen	Oxygen	Phosphorous	Galactose	✓		✓		Phospholipid	✓		✓	✓	RNA	✓	✓	✓	✓	Sucrose	✓		✓			
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09.2	Correct answer for 2 marks, 0.8376308/0.84/0.8 (hours);; Accept for 1 mark, Evidence of 4×10^6 and 3×10^9 (written in any format, for correct readings from graph) OR Evidence of 9.550746785 (correct number of generations) OR Evidence of 1.1938443348 (correct generations/ hour) OR Evidence of 50.26 (correct generation time in minutes) _____ Incorrect reading of graph, 3×10^6 and 2×10^9 Accept for 1 mark, calculation carried out correctly Evidence of 9.380821784 (correct calculation of number of generations)	2	Accept correct rounding to any number of decimal places

	OR Evidence of 1.172602723 (correct calculation of generations/ hour) OR Evidence of 51.16822503 (correct calculation of generation time in minutes) OR Evidence of 0.8528037505 (correct calculation of generation time in hours) _____ Incorrect reading of graph, $10^{6.4}$ and $10^{9.3}$ OR $10^{6.3}$ and $10^{9.2}$ Accept for 1 mark, calculation carried out correctly Evidence of 9.633591475 (correct calculation of number of generations) OR Evidence of 1.204198934 (correct calculation of generations/ hour) OR Evidence of 49.82565445 (correct calculation of generation time in minutes) OR Evidence of 0.8304275742 (correct calculation of generation time in hours) OR Evidence of 0.83, with no other working		
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Question	Marking Guidance	Mark	Comments
09.3	1. (Model) Q and (Name) Semi-conservative (replication); Explanation 2. (Model) P (is unsupported because) There should be two peaks in generation 1 OR (Only) one peak is shown in generation 1 OR There should be 3:1 (ratio) of peaks in generation 2 OR There should not be an intermediate/¹⁵N ¹⁴N peak in generation 1/2/3 OR The original/generation 0/¹⁵N peak should be in generation 1/2/3; 3. (Model) R (is unsupported because) There should be >2 peaks in generation 2/3 OR There should be one wide/overlapping peak in generation 3;	3	Accept answers 2. and 3. in either order Accept for 'peak', density OR distribution 3. Accept for '>2', many OR several